Executive Engineering Report

E2E MultiUser Test Project 1787123629160

ELECTRICAL DESIGN & INFRASTRUCTURE REPORT

Prepared in accordance with IEC 60364 & BS 7671 Electrical Regulations

Report Ref: PRJ-921988
Date: 2026-08-19
Client NameAl-Hamra Holdings
ConsultantKhatib & Alami
Lead EngineerLead Electrical Engineer

1. System Electrical Calculations Summary

Total Max Demand608.1 kW
Calculated Current1040.6 A
System Voltage400V 3-Phase
Utility Transformer1000 kVA (400V)

2. Project Distribution Hierarchy & Infrastructure

Building / Structure	Floors	Main Incomer Breaker	Main Feeder Cable	Distribution Panels	Max Demand (kW / Amps)
tower A	10 Floors	1000A ACB	2 × (4C × 240 mm²)	10 Sub-Panels	480.3 kW (823.6A)
Tower B	7 Floors	250A MCCB	4C × 95 mm²	9 Sub-Panels	127.8 kW (217.0A)

3. Document Revisions History

Rev	Date	Description	Prepared By
R0	2026-08-19	Initial engineering release	Lead Electrical Engineer

Engineering Sign-Off

DesignedLead Electrical Engineer
QC CheckQC Dept
StatusAPPROVED

E2E MultiUser Test Project 1787123629160

LOAD ANALYSIS & PHASE BALANCING SCHEDULE

Prepared in accordance with IEC 60364 & BS 7671 Electrical Regulations

Ref: PRJ-921988
Date: 2026-08-19

Load Analysis & Phase Balancing Schedule

Standard: IEC
Total Connected1216.3 kW
Max Demand608.1 kW
Phase Balance (L1/L2/L3)1041A / 1041A / 1016A
Max Phase Current1040.6 A

#	Building	Floor	Load / Circuit	Type	Conn. (kW)	DF	Demand (kW)	Phase	L1 (A)	L2 (A)	L3 (A)	PF
1	tower A	F3	test (F3-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
2	tower A	F3	lux1 (F3-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
3	tower A	F3	test (F3-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
4	tower A	F3	test (F3-D)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
5	tower A	F3	test (F3-E)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
6	tower A	F3	lux1 (F3-F)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
7	tower A	F3	test (F3-G)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
8	tower A	F3	test (F3-H)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
9	tower A	F4	test (F4-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
Total System Max Demand:					1216.3 kW	—	608.1 kW	3Φ	1040.6A	1040.6A	1016.5A	0.85

#	Building	Floor	Load / Circuit	Type	Conn. (kW)	DF	Demand (kW)	Phase	L1 (A)	L2 (A)	L3 (A)	PF
10	tower A	F4	lux1 (F4-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
11	tower A	F4	test (F4-C)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
12	tower A	F4	test (F4-D)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
13	tower A	F4	test (F4-E)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
14	tower A	F4	lux1 (F4-F)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
15	tower A	F4	test (F4-G)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
16	tower A	F4	test (F4-H)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
17	tower A	F5	test (F5-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
18	tower A	F5	lux1 (F5-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
19	tower A	F5	test (F5-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
20	tower A	F5	test (F5-D)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
21	tower A	F5	test (F5-E)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
22	tower A	F5	lux1 (F5-F)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
23	tower A	F5	test (F5-G)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
24	tower A	F5	test (F5-H)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
25	tower A	F6	test (F6-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
26	tower A	F6	lux1 (F6-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
27	tower A	F6	test (F6-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
28	tower A	F6	test (F6-D)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
29	tower A	F6	test (F6-E)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
30	tower A	F6	test (F6-F)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
31	tower A	F6	test (F6-G)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
32	tower A	F6	lux1 (F6-H)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
33	tower A	F9	test (F9-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
34	tower A	F9	lux1 (F9-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
35	tower A	F9	test (F9-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
36	tower A	F9	test (F9-D)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
37	tower A	F9	test (F9-E)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
38	tower A	F9	lux1 (F9-F)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
Total System Max Demand:					1216.3 kW	—	608.1 kW	3Φ	1040.6A	1040.6A	1016.5A	0.85

#	Building	Floor	Load / Circuit	Type	Conn. (kW)	DF	Demand (kW)	Phase	L1 (A)	L2 (A)	L3 (A)	PF
39	tower A	F9	test (F9-G)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
40	tower A	F9	test (F9-H)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
41	tower A	F10	test (F10-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
42	tower A	F10	lux1 (F10-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
43	tower A	F10	test (F10-C)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
44	tower A	F10	test (F10-D)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
45	tower A	F10	lux1 (F10-E)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
46	tower A	F10	test (F10-F)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
47	tower A	F10	test (F10-G)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
48	tower A	F10	test (F10-H)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
49	tower A	F2	test (F2-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
50	tower A	F2	lux1 (F2-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
51	tower A	F2	test (F2-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
52	tower A	F2	test (F2-D)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
53	tower A	F2	lux1 (F2-E)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
54	tower A	F2	test (F2-F)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
55	tower A	F2	test (F2-G)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
56	tower A	F2	test (F2-H)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
57	tower A	F7	lux1 (F7-A)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
58	tower A	F7	test (F7-B)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
59	tower A	F7	lux1 (F7-C)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
60	tower A	F7	test (F7-D)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
61	tower A	F7	test (F7-E)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
62	tower A	F7	test (F7-F)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
63	tower A	F7	test (F7-G)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
64	tower A	F7	test (F7-H)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
65	tower A	F8	test (F8-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
66	tower A	F8	lux1 (F8-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
67	tower A	F8	test (F8-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
Total System Max Demand:					1216.3 kW	—	608.1 kW	3Φ	1040.6A	1040.6A	1016.5A	0.85

#	Building	Floor	Load / Circuit	Type	Conn. (kW)	DF	Demand (kW)	Phase	L1 (A)	L2 (A)	L3 (A)	PF
68	tower A	F8	test (F8-D)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
69	tower A	F8	test (F8-E)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
70	tower A	F8	test (F8-F)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
71	tower A	F8	lux1 (F8-G)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
72	tower A	F8	test (F8-H)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
73	tower A	F1	test (F1-A)	APARTMENT	16.6	0.50	8.3	3Φ	14.1	14.1	14.1	0.85
74	tower A	F1	lux1 (F1-B)	APARTMENT	15.0	0.50	7.5	3Φ	12.7	12.7	12.7	0.85
75	tower A	F1	test (F1-C)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
76	tower A	F1	test (F1-D)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
77	Tower B	F6	test (F6-A)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
78	Tower B	F6	test2 (F6-B)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
79	Tower B	F6	test3 (F6-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
80	Tower B	F7	test (F7-A)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
81	Tower B	F7	test2 (F7-B)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
82	Tower B	F7	test3 (F7-C)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
83	Tower B	F8	test (F8-A)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
84	Tower B	F8	test2 (F8-B)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
85	Tower B	F8	test3 (F8-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
86	Tower B	F9	test (F9-A)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
87	Tower B	F9	test2 (F9-B)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
88	Tower B	F9	test3 (F9-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
89	Tower B	F4	test (F4-A)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
90	Tower B	F4	test2 (F4-B)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
91	Tower B	F4	test3 (F4-C)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
92	Tower B	F5	test (F5-A)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
93	Tower B	F5	test2 (F5-B)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
94	Tower B	F5	test3 (F5-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
95	Tower B	F1	test3 (F1-A)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
96	Tower B	F1	test (F1-B)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
Total System Max Demand:					1216.3 kW	—	608.1 kW	3Φ	1040.6A	1040.6A	1016.5A	0.85

#	Building	Floor	Load / Circuit	Type	Conn. (kW)	DF	Demand (kW)	Phase	L1 (A)	L2 (A)	L3 (A)	PF
97	Tower B	F1	test2 (F1-C)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
98	Tower B	F2	test (F2-A)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
99	Tower B	F2	test2 (F2-B)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
100	Tower B	F2	test3 (F2-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
101	Tower B	F3	test (F3-A)	APARTMENT	9.5	0.50	4.7	1Φ	24.1	—	—	0.85
102	Tower B	F3	test2 (F3-B)	APARTMENT	9.5	0.50	4.7	1Φ	—	24.1	—	0.85
103	Tower B	F3	test3 (F3-C)	APARTMENT	9.5	0.50	4.7	1Φ	—	—	24.1	0.85
Total System Max Demand:					1216.3 kW	—	608.1 kW	3Φ	1040.6A	1040.6A	1016.5A	0.85

E2E MultiUser Test Project 1787123629160

MAIN DISTRIBUTION BOARD (MDB) FEEDER SCHEDULE

Prepared in accordance with IEC 60364 & BS 7671 Electrical Regulations

Ref: PRJ-921988
Date: 2026-08-19

Main Distribution Board (MDB) Schedule

Standard: IEC

#	Building	Floor	Feeder Description	Type	Demand (kW)	Current (A)	Protection (In)	Feeder Cable	Iz (A)
1	tower A	—	Main Incomer	ACB	480.3	815.5	1000A	2 × 240 mm²	1000
2	tower A	F3	F3 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
3	tower A	F3	F3 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
4	tower A	F3	F3 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
5	tower A	F3	F3 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
6	tower A	F3	F3 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
7	tower A	F3	F3 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
8	tower A	F3	F3 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
9	tower A	F3	F3 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
10	tower A	F4	F4 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
11	tower A	F4	F4 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
12	tower A	F4	F4 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
13	tower A	F4	F4 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33

#	Building	Floor	Feeder Description	Type	Demand (kW)	Current (A)	Protection (In)	Feeder Cable	Iz (A)
14	tower A	F4	F4 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
15	tower A	F4	F4 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
16	tower A	F4	F4 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
17	tower A	F4	F4 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
18	tower A	F5	F5 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
19	tower A	F5	F5 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
20	tower A	F5	F5 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
21	tower A	F5	F5 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
22	tower A	F5	F5 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
23	tower A	F5	F5 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
24	tower A	F5	F5 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
25	tower A	F5	F5 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
26	tower A	F6	F6 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
27	tower A	F6	F6 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
28	tower A	F6	F6 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
29	tower A	F6	F6 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
30	tower A	F6	F6 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
31	tower A	F6	F6 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
32	tower A	F6	F6 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
33	tower A	F6	F6 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
34	tower A	F9	F9 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
35	tower A	F9	F9 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
36	tower A	F9	F9 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
37	tower A	F9	F9 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
38	tower A	F9	F9 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
39	tower A	F9	F9 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
40	tower A	F9	F9 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
41	tower A	F9	F9 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
42	tower A	F10	F10 – SMDB	SMDB	60.0	101.9	160A	50 mm²	179
43	tower A	F2	F2 – SMDB	SMDB	60.0	101.9	160A	50 mm²	179

#	Building	Floor	Feeder Description	Type	Demand (kW)	Current (A)	Protection (In)	Feeder Cable	Iz (A)
44	tower A	F7	F7 – SMDB	SMDB	60.0	101.9	160A	50 mm²	179
45	tower A	F8	F8 – SMDB	SMDB	60.0	101.9	160A	50 mm²	179
46	tower A	F1	F1 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
47	tower A	F1	F1 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
48	tower A	F1	F1 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
49	tower A	F1	F1 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
50	tower A	F10	F10 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
51	tower A	F10	F10 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
52	tower A	F10	F10 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
53	tower A	F10	F10 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
54	tower A	F10	F10 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
55	tower A	F10	F10 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
56	tower A	F10	F10 – test	APARTMENT	14.2	24.1	25A	4 mm²	45
57	tower A	F10	F10 – test	APARTMENT	14.2	24.1	25A	4 mm²	45
58	tower A	F2	F2 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
59	tower A	F2	F2 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
60	tower A	F2	F2 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
61	tower A	F2	F2 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
62	tower A	F2	F2 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
63	tower A	F2	F2 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
64	tower A	F2	F2 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
65	tower A	F2	F2 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
66	tower A	F7	F7 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
67	tower A	F7	F7 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
68	tower A	F7	F7 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
69	tower A	F7	F7 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
70	tower A	F7	F7 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
71	tower A	F7	F7 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
72	tower A	F7	F7 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
73	tower A	F7	F7 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33

#	Building	Floor	Feeder Description	Type	Demand (kW)	Current (A)	Protection (In)	Feeder Cable	Iz (A)
74	tower A	F8	F8 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
75	tower A	F8	F8 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
76	tower A	F8	F8 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
77	tower A	F8	F8 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
78	tower A	F8	F8 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
79	tower A	F8	F8 – test	APARTMENT	8.3	14.1	16A	1.5 mm²	22
80	tower A	F8	F8 – lux1	APARTMENT	7.5	12.7	16A	1.5 mm²	22
81	tower A	F8	F8 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
82	Tower B	—	Main Incomer	MCCB	127.8	217.0	250A	95 mm²	278
83	Tower B	F6	F6 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
84	Tower B	F6	F6 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33
85	Tower B	F6	F6 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33
86	Tower B	F7	F7 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
87	Tower B	F7	F7 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33
88	Tower B	F7	F7 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33
89	Tower B	F8	F8 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
90	Tower B	F8	F8 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33
91	Tower B	F8	F8 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33
92	Tower B	F9	F9 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
93	Tower B	F9	F9 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33
94	Tower B	F9	F9 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33
95	Tower B	F4	F4 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
96	Tower B	F4	F4 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33
97	Tower B	F4	F4 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33
98	Tower B	F5	F5 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
99	Tower B	F5	F5 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33
100	Tower B	F5	F5 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33
101	Tower B	F1	F1 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33
102	Tower B	F1	F1 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
103	Tower B	F1	F1 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33

#	Building	Floor	Feeder Description	Type	Demand (kW)	Current (A)	Protection (In)	Feeder Cable	Iz (A)
104	Tower B	F2	F2 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
105	Tower B	F2	F2 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33
106	Tower B	F2	F2 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33
107	Tower B	F3	F3 – test	APARTMENT	14.2	24.1	25A	2.5 mm²	33
108	Tower B	F3	F3 – test2	APARTMENT	14.2	24.1	25A	2.5 mm²	33
109	Tower B	F3	F3 – test3	APARTMENT	14.2	24.1	25A	2.5 mm²	33

E2E MultiUser Test Project 1787123629160

CABLE SIZING & INSTALLATION SCHEDULE

Prepared in accordance with IEC 60364 & BS 7671 Electrical Regulations

Ref: PRJ-921988
Date: 2026-08-19

Cable Sizing, Derating & Installation Schedule

Standard: IEC 60364-5-52 / BS 7671

#	Building	Floor	Circuit / Feeder	Phase	Design Ib (A)	Breaker (In)	Cable Size	Install Method	Insulation & Material
1	tower A	MDB	tower A – Main Incomer Feeder	3Φ	815.5 A	1000A	2 × (4C × 240 mm²)	Method E	XLPE / Cu
2	tower A	F3	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
3	tower A	F3	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
4	tower A	F3	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
5	tower A	F3	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
6	tower A	F3	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
7	tower A	F3	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
8	tower A	F3	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
9	tower A	F3	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
10	tower A	F4	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
11	tower A	F4	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
12	tower A	F4	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
13	tower A	F4	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu

#	Building	Floor	Circuit / Feeder	Phase	Design Ib (A)	Breaker (In)	Cable Size	Install Method	Insulation & Material
14	tower A	F4	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
15	tower A	F4	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
16	tower A	F4	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
17	tower A	F4	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
18	tower A	F5	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
19	tower A	F5	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
20	tower A	F5	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
21	tower A	F5	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
22	tower A	F5	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
23	tower A	F5	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
24	tower A	F5	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
25	tower A	F5	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
26	tower A	F6	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
27	tower A	F6	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
28	tower A	F6	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
29	tower A	F6	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
30	tower A	F6	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
31	tower A	F6	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
32	tower A	F6	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
33	tower A	F6	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
34	tower A	F9	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
35	tower A	F9	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
36	tower A	F9	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
37	tower A	F9	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
38	tower A	F9	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
39	tower A	F9	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
40	tower A	F9	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
41	tower A	F9	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
42	tower A	F10	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
43	tower A	F10	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu

#	Building	Floor	Circuit / Feeder	Phase	Design Ib (A)	Breaker (In)	Cable Size	Install Method	Insulation & Material
44	tower A	F10	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
45	tower A	F10	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
46	tower A	F10	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
47	tower A	F10	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
48	tower A	F10	test	1Φ	24.1 A	25A	4 mm²	Method C	XLPE / Cu
49	tower A	F10	test	1Φ	24.1 A	25A	4 mm²	Method C	XLPE / Cu
50	tower A	F2	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
51	tower A	F2	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
52	tower A	F2	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
53	tower A	F2	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
54	tower A	F2	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
55	tower A	F2	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
56	tower A	F2	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
57	tower A	F2	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
58	tower A	F7	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
59	tower A	F7	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
60	tower A	F7	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
61	tower A	F7	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
62	tower A	F7	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
63	tower A	F7	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
64	tower A	F7	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
65	tower A	F7	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
66	tower A	F8	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
67	tower A	F8	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
68	tower A	F8	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
69	tower A	F8	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
70	tower A	F8	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
71	tower A	F8	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
72	tower A	F8	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
73	tower A	F8	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu

#	Building	Floor	Circuit / Feeder	Phase	Design Ib (A)	Breaker (In)	Cable Size	Install Method	Insulation & Material
74	tower A	F1	test	3Φ	14.1 A	16A	1.5 mm²	Method C	XLPE / Cu
75	tower A	F1	lux1	3Φ	12.7 A	16A	1.5 mm²	Method C	XLPE / Cu
76	tower A	F1	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
77	tower A	F1	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
78	Tower B	MDB	Tower B – Main Incomer Feeder	3Φ	217.0 A	250A	4C × 95 mm²	Method E	XLPE / Cu
79	Tower B	F6	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
80	Tower B	F6	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
81	Tower B	F6	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
82	Tower B	F7	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
83	Tower B	F7	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
84	Tower B	F7	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
85	Tower B	F8	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
86	Tower B	F8	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
87	Tower B	F8	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
88	Tower B	F9	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
89	Tower B	F9	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
90	Tower B	F9	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
91	Tower B	F4	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
92	Tower B	F4	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
93	Tower B	F4	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
94	Tower B	F5	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
95	Tower B	F5	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
96	Tower B	F5	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
97	Tower B	F1	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
98	Tower B	F1	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
99	Tower B	F1	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
100	Tower B	F2	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
101	Tower B	F2	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
102	Tower B	F2	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
103	Tower B	F3	test	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu

#	Building	Floor	Circuit / Feeder	Phase	Design Ib (A)	Breaker (In)	Cable Size	Install Method	Insulation & Material
104	Tower B	F3	test2	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu
105	Tower B	F3	test3	1Φ	24.1 A	25A	2.5 mm²	Method C	XLPE / Cu

E2E MultiUser Test Project 1787123629160

CIRCUIT BREAKERS & SELECTIVITY PROTECTION SCHEDULE

Prepared in accordance with IEC 60364 & BS 7671 Electrical Regulations

Ref: PRJ-921988
Date: 2026-08-19

Circuit Breakers & Protection Coordination Schedule

Standard: IEC 60947-2 / IEC 60898-1

INCOMER(2 devices)

Feeder Description	Upstream Parent	Floor	Ib (A)	Rating (In)	Breaker Model	Cable	Isc (kA)	Selectivity	Thermal Withstand
tower A – Main Incomer	Utility / Transformer Supply	MDB	815.5 A	1000A	Schneider Masterpact MTZ2 MTZ2 H1 1000	240 mm²	20.99 kA	FULL	I²t ≤ k²S² (OK)
Tower B – Main Incomer	Utility / Transformer Supply	MDB	217.0 A	250A	Schneider ComPacT NSX250 NSX250F MicroLogic 2.2	95 mm²	8.02 kA	FULL	I²t ≤ k²S² (OK)

APARTMENT(103 devices)

Feeder Description	Upstream Parent	Floor	Ib (A)	Rating (In)	Breaker Model	Cable	Isc (kA)	Selectivity	Thermal Withstand
F3 – test	Main Incomer	F3	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	I²t ≤ k²S² (OK)
F3 – lux1	Main Incomer	F3	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	I²t ≤ k²S² (OK)

Feeder Description	Upstream Parent	Floor	Ib (A)	Rating (In)	Breaker Model	Cable	Isc (kA)	Selectivity	Thermal Withstand
F3 – test	Main Incomer	F3	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F3 – test	Main Incomer	F3	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F3 – test	Main Incomer	F3	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F3 – lux1	Main Incomer	F3	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	$I^2t \le k^2S^2$ (OK)
F3 – test	Main Incomer	F3	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F3 – test	Main Incomer	F3	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test	Main Incomer	F4	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – lux1	Main Incomer	F4	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test	Main Incomer	F4	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test	Main Incomer	F4	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test	Main Incomer	F4	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – lux1	Main Incomer	F4	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test	Main Incomer	F4	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test	Main Incomer	F4	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – test	Main Incomer	F5	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – lux1	Main Incomer	F5	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – test	Main Incomer	F5	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – test	Main Incomer	F5	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – test	Main Incomer	F5	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – lux1	Main Incomer	F5	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – test	Main Incomer	F5	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – test	Main Incomer	F5	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test	Main Incomer	F6	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – lux1	Main Incomer	F6	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test	Main Incomer	F6	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test	Main Incomer	F6	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test	Main Incomer	F6	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test	Main Incomer	F6	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test	Main Incomer	F6	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – lux1	Main Incomer	F6	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	$I^2t \le k^2S^2$ (OK)

Feeder Description	Upstream Parent	Floor	Ib (A)	Rating (In)	Breaker Model	Cable	Isc (kA)	Selectivity	Thermal Withstand
F9 – test	Main Incomer	F9	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	I²t ≤ k²S² (OK)
F9 – lux1	Main Incomer	F9	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	I²t ≤ k²S² (OK)
F9 – test	Main Incomer	F9	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	I²t ≤ k²S² (OK)
F9 – test	Main Incomer	F9	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	I²t ≤ k²S² (OK)
F9 – test	Main Incomer	F9	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	I²t ≤ k²S² (OK)
F9 – lux1	Main Incomer	F9	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	I²t ≤ k²S² (OK)
F9 – test	Main Incomer	F9	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	I²t ≤ k²S² (OK)
F9 – test	Main Incomer	F9	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	I²t ≤ k²S² (OK)
F1 – test	Main Incomer	F1	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	I²t ≤ k²S² (OK)
F1 – lux1	Main Incomer	F1	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	I²t ≤ k²S² (OK)
F1 – test	Main Incomer	F1	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	I²t ≤ k²S² (OK)
F1 – test	Main Incomer	F1	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.64 kA	FULL	I²t ≤ k²S² (OK)
F10 – test	F10 – SMDB	F10	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.71 kA	FULL	I²t ≤ k²S² (OK)
F10 – lux1	F10 – SMDB	F10	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.92 kA	FULL	I²t ≤ k²S² (OK)
F10 – test	F10 – SMDB	F10	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.49 kA	FULL	I²t ≤ k²S² (OK)
F10 – test	F10 – SMDB	F10	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.47 kA	FULL	I²t ≤ k²S² (OK)
F10 – lux1	F10 – SMDB	F10	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.28 kA	FULL	I²t ≤ k²S² (OK)
F10 – test	F10 – SMDB	F10	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.28 kA	FULL	I²t ≤ k²S² (OK)
F10 – test	F10 – SMDB	F10	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	4 mm²	0.36 kA	FULL	I²t ≤ k²S² (OK)
F10 – test	F10 – SMDB	F10	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	4 mm²	0.36 kA	FULL	I²t ≤ k²S² (OK)
F2 – test	F2 – SMDB	F2	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.74 kA	FULL	I²t ≤ k²S² (OK)
F2 – lux1	F2 – SMDB	F2	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.97 kA	FULL	I²t ≤ k²S² (OK)
F2 – test	F2 – SMDB	F2	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.50 kA	FULL	I²t ≤ k²S² (OK)
F2 – test	F2 – SMDB	F2	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.49 kA	FULL	I²t ≤ k²S² (OK)
F2 – lux1	F2 – SMDB	F2	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.97 kA	FULL	I²t ≤ k²S² (OK)
F2 – test	F2 – SMDB	F2	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.82 kA	FULL	I²t ≤ k²S² (OK)
F2 – test	F2 – SMDB	F2	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.97 kA	FULL	I²t ≤ k²S² (OK)
F2 – test	F2 – SMDB	F2	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.82 kA	FULL	I²t ≤ k²S² (OK)
F7 – lux1	F7 – SMDB	F7	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.38 kA	FULL	I²t ≤ k²S² (OK)
F7 – test	F7 – SMDB	F7	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.74 kA	FULL	I²t ≤ k²S² (OK)

Feeder Description	Upstream Parent	Floor	Ib (A)	Rating (In)	Breaker Model	Cable	Isc (kA)	Selectivity	Thermal Withstand
F7 – lux1	F7 – SMDB	F7	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.96 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – test	F7 – SMDB	F7	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.50 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – test	F7 – SMDB	F7	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.48 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – test	F7 – SMDB	F7	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.32 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – test	F7 – SMDB	F7	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.38 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – test	F7 – SMDB	F7	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.32 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test	F8 – SMDB	F8	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.76 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – lux1	F8 – SMDB	F8	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	1.00 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test	F8 – SMDB	F8	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.51 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test	F8 – SMDB	F8	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.49 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test	F8 – SMDB	F8	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.29 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test	F8 – SMDB	F8	14.1 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.34 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – lux1	F8 – SMDB	F8	12.7 A	16A	Schneider Acti9 iC60 iC60N 3P 16A C	1.5 mm²	0.34 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test	F8 – SMDB	F8	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.29 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test	Main Incomer	F6	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.36 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test2	Main Incomer	F6	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.36 kA	FULL	$I^2t \le k^2S^2$ (OK)
F6 – test3	Main Incomer	F6	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.36 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – test	Main Incomer	F7	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.31 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – test2	Main Incomer	F7	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.31 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – test3	Main Incomer	F7	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.31 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test	Main Incomer	F8	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.28 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test2	Main Incomer	F8	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.28 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – test3	Main Incomer	F8	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.28 kA	FULL	$I^2t \le k^2S^2$ (OK)
F9 – test	Main Incomer	F9	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.25 kA	FULL	$I^2t \le k^2S^2$ (OK)
F9 – test2	Main Incomer	F9	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.25 kA	FULL	$I^2t \le k^2S^2$ (OK)
F9 – test3	Main Incomer	F9	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.25 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test	Main Incomer	F4	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.49 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test2	Main Incomer	F4	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.49 kA	FULL	$I^2t \le k^2S^2$ (OK)
F4 – test3	Main Incomer	F4	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.49 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – test	Main Incomer	F5	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.41 kA	FULL	$I^2t \le k^2S^2$ (OK)

Feeder Description	Upstream Parent	Floor	Ib (A)	Rating (In)	Breaker Model	Cable	Isc (kA)	Selectivity	Thermal Withstand
F5 – test2	Main Incomer	F5	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.41 kA	FULL	$I^2t \le k^2S^2$ (OK)
F5 – test3	Main Incomer	F5	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.41 kA	FULL	$I^2t \le k^2S^2$ (OK)
F1 – test3	Main Incomer	F1	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	1.13 kA	FULL	$I^2t \le k^2S^2$ (OK)
F1 – test	Main Incomer	F1	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	1.13 kA	FULL	$I^2t \le k^2S^2$ (OK)
F1 – test2	Main Incomer	F1	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	1.13 kA	FULL	$I^2t \le k^2S^2$ (OK)
F2 – test	Main Incomer	F2	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.79 kA	FULL	$I^2t \le k^2S^2$ (OK)
F2 – test2	Main Incomer	F2	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.79 kA	FULL	$I^2t \le k^2S^2$ (OK)
F2 – test3	Main Incomer	F2	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.79 kA	FULL	$I^2t \le k^2S^2$ (OK)
F3 – test	Main Incomer	F3	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.61 kA	FULL	$I^2t \le k^2S^2$ (OK)
F3 – test2	Main Incomer	F3	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.61 kA	FULL	$I^2t \le k^2S^2$ (OK)
F3 – test3	Main Incomer	F3	24.1 A	25A	Schneider Acti9 iC60 iC60N 1P 25A C	2.5 mm²	0.61 kA	FULL	$I^2t \le k^2S^2$ (OK)

SMDB(4 devices)

Feeder Description	Upstream Parent	Floor	Ib (A)	Rating (In)	Breaker Model	Cable	Isc (kA)	Selectivity	Thermal Withstand
F10 – SMDB	Main Incomer	F10	101.9 A	160A	Schneider ComPacT NSX160 NSX160F MicroLogic 2.2	50 mm²	7.81 kA	FULL	$I^2t \le k^2S^2$ (OK)
F2 – SMDB	Main Incomer	F2	101.9 A	160A	Schneider ComPacT NSX160 NSX160F MicroLogic 2.2	50 mm²	13.04 kA	FULL	$I^2t \le k^2S^2$ (OK)
F7 – SMDB	Main Incomer	F7	101.9 A	160A	Schneider ComPacT NSX160 NSX160F MicroLogic 2.2	50 mm²	11.57 kA	FULL	$I^2t \le k^2S^2$ (OK)
F8 – SMDB	Main Incomer	F8	101.9 A	160A	Schneider ComPacT NSX160 NSX160F MicroLogic 2.2	50 mm²	19.41 kA	FULL	$I^2t \le k^2S^2$ (OK)

E2E MultiUser Test Project 1787123629160

VOLTAGE DROP & COMPLIANCE ANALYSIS SCHEDULE

Prepared in accordance with IEC 60364 & BS 7671 Electrical Regulations

MCGI GLASS

Ref: PRJ-921988
Date: 2026-08-19

Voltage Drop & Compliance Analysis Schedule

Standard: IEC 60364-5-52

#	Building	Floor	Circuit / Feeder	Ib (A)	Cable Size	Length (m)	Voltage Drop (ΔV %)	Compliance Status
1	tower A	F3	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
2	tower A	F3	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
3	tower A	F3	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
4	tower A	F3	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
5	tower A	F3	test	24.1 A	2.5 mm²	20 m	0.10 %	PASS (<= 4%)
6	tower A	F3	lux1	12.7 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
7	tower A	F3	test	24.1 A	2.5 mm²	20 m	0.10 %	PASS (<= 4%)
8	tower A	F3	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
9	tower A	F4	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
10	tower A	F4	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
11	tower A	F4	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
12	tower A	F4	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
13	tower A	F4	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)

#	Building	Floor	Circuit / Feeder	Ib (A)	Cable Size	Length (m)	Voltage Drop (ΔV %)	Compliance Status
14	tower A	F4	lux1	12.7 A	1.5 mm²	25 m	0.10 %	PASS (<= 4%)
15	tower A	F4	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
16	tower A	F4	test	14.1 A	1.5 mm²	25 m	0.10 %	PASS (<= 4%)
17	tower A	F5	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
18	tower A	F5	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
19	tower A	F5	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
20	tower A	F5	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
21	tower A	F5	test	24.1 A	2.5 mm²	30 m	0.10 %	PASS (<= 4%)
22	tower A	F5	lux1	12.7 A	1.5 mm²	30 m	0.10 %	PASS (<= 4%)
23	tower A	F5	test	24.1 A	2.5 mm²	30 m	0.10 %	PASS (<= 4%)
24	tower A	F5	test	14.1 A	1.5 mm²	30 m	0.10 %	PASS (<= 4%)
25	tower A	F6	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
26	tower A	F6	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
27	tower A	F6	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
28	tower A	F6	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
29	tower A	F6	test	14.1 A	1.5 mm²	35 m	0.10 %	PASS (<= 4%)
30	tower A	F6	test	24.1 A	2.5 mm²	35 m	0.10 %	PASS (<= 4%)
31	tower A	F6	test	24.1 A	2.5 mm²	35 m	0.10 %	PASS (<= 4%)
32	tower A	F6	lux1	12.7 A	1.5 mm²	35 m	0.10 %	PASS (<= 4%)
33	tower A	F9	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
34	tower A	F9	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
35	tower A	F9	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
36	tower A	F9	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
37	tower A	F9	test	24.1 A	2.5 mm²	50 m	0.10 %	PASS (<= 4%)
38	tower A	F9	lux1	12.7 A	1.5 mm²	50 m	0.10 %	PASS (<= 4%)
39	tower A	F9	test	24.1 A	2.5 mm²	50 m	0.10 %	PASS (<= 4%)
40	tower A	F9	test	14.1 A	1.5 mm²	50 m	0.10 %	PASS (<= 4%)
41	tower A	F10	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
42	tower A	F10	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
43	tower A	F10	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)

#	Building	Floor	Circuit / Feeder	Ib (A)	Cable Size	Length (m)	Voltage Drop (ΔV %)	Compliance Status
44	tower A	F10	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
45	tower A	F10	lux1	12.7 A	1.5 mm²	55 m	0.10 %	PASS (<= 4%)
46	tower A	F10	test	14.1 A	1.5 mm²	55 m	0.10 %	PASS (<= 4%)
47	tower A	F10	test	24.1 A	4 mm²	55 m	0.10 %	PASS (<= 4%)
48	tower A	F10	test	24.1 A	4 mm²	55 m	0.10 %	PASS (<= 4%)
49	tower A	F2	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
50	tower A	F2	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
51	tower A	F2	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
52	tower A	F2	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
53	tower A	F2	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
54	tower A	F2	test	24.1 A	2.5 mm²	15 m	0.10 %	PASS (<= 4%)
55	tower A	F2	test	14.1 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
56	tower A	F2	test	24.1 A	2.5 mm²	15 m	0.10 %	PASS (<= 4%)
57	tower A	F7	lux1	12.7 A	1.5 mm²	40 m	0.10 %	PASS (<= 4%)
58	tower A	F7	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
59	tower A	F7	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
60	tower A	F7	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
61	tower A	F7	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
62	tower A	F7	test	24.1 A	2.5 mm²	40 m	0.10 %	PASS (<= 4%)
63	tower A	F7	test	14.1 A	1.5 mm²	40 m	0.10 %	PASS (<= 4%)
64	tower A	F7	test	24.1 A	2.5 mm²	40 m	0.10 %	PASS (<= 4%)
65	tower A	F8	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)
66	tower A	F8	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
67	tower A	F8	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
68	tower A	F8	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
69	tower A	F8	test	24.1 A	2.5 mm²	45 m	0.10 %	PASS (<= 4%)
70	tower A	F8	test	14.1 A	1.5 mm²	45 m	0.10 %	PASS (<= 4%)
71	tower A	F8	lux1	12.7 A	1.5 mm²	45 m	0.10 %	PASS (<= 4%)
72	tower A	F8	test	24.1 A	2.5 mm²	45 m	0.10 %	PASS (<= 4%)
73	tower A	F1	test	14.1 A	1.5 mm²	20 m	0.10 %	PASS (<= 4%)

#	Building	Floor	Circuit / Feeder	Ib (A)	Cable Size	Length (m)	Voltage Drop (ΔV %)	Compliance Status
74	tower A	F1	lux1	12.7 A	1.5 mm²	15 m	0.10 %	PASS (<= 4%)
75	tower A	F1	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
76	tower A	F1	test	24.1 A	2.5 mm²	26 m	0.10 %	PASS (<= 4%)
77	Tower B	F6	test	24.1 A	2.5 mm²	35 m	0.10 %	PASS (<= 4%)
78	Tower B	F6	test2	24.1 A	2.5 mm²	35 m	0.10 %	PASS (<= 4%)
79	Tower B	F6	test3	24.1 A	2.5 mm²	35 m	0.10 %	PASS (<= 4%)
80	Tower B	F7	test	24.1 A	2.5 mm²	40 m	0.10 %	PASS (<= 4%)
81	Tower B	F7	test2	24.1 A	2.5 mm²	40 m	0.10 %	PASS (<= 4%)
82	Tower B	F7	test3	24.1 A	2.5 mm²	40 m	0.10 %	PASS (<= 4%)
83	Tower B	F8	test	24.1 A	2.5 mm²	45 m	0.10 %	PASS (<= 4%)
84	Tower B	F8	test2	24.1 A	2.5 mm²	45 m	0.10 %	PASS (<= 4%)
85	Tower B	F8	test3	24.1 A	2.5 mm²	45 m	0.10 %	PASS (<= 4%)
86	Tower B	F9	test	24.1 A	2.5 mm²	50 m	0.10 %	PASS (<= 4%)
87	Tower B	F9	test2	24.1 A	2.5 mm²	50 m	0.10 %	PASS (<= 4%)
88	Tower B	F9	test3	24.1 A	2.5 mm²	50 m	0.10 %	PASS (<= 4%)
89	Tower B	F4	test	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
90	Tower B	F4	test2	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
91	Tower B	F4	test3	24.1 A	2.5 mm²	25 m	0.10 %	PASS (<= 4%)
92	Tower B	F5	test	24.1 A	2.5 mm²	30 m	0.10 %	PASS (<= 4%)
93	Tower B	F5	test2	24.1 A	2.5 mm²	30 m	0.10 %	PASS (<= 4%)
94	Tower B	F5	test3	24.1 A	2.5 mm²	30 m	0.10 %	PASS (<= 4%)
95	Tower B	F1	test3	24.1 A	2.5 mm²	10 m	0.10 %	PASS (<= 4%)
96	Tower B	F1	test	24.1 A	2.5 mm²	10 m	0.10 %	PASS (<= 4%)
97	Tower B	F1	test2	24.1 A	2.5 mm²	10 m	0.10 %	PASS (<= 4%)
98	Tower B	F2	test	24.1 A	2.5 mm²	15 m	0.10 %	PASS (<= 4%)
99	Tower B	F2	test2	24.1 A	2.5 mm²	15 m	0.10 %	PASS (<= 4%)
100	Tower B	F2	test3	24.1 A	2.5 mm²	15 m	0.10 %	PASS (<= 4%)
101	Tower B	F3	test	24.1 A	2.5 mm²	20 m	0.10 %	PASS (<= 4%)
102	Tower B	F3	test2	24.1 A	2.5 mm²	20 m	0.10 %	PASS (<= 4%)
103	Tower B	F3	test3	24.1 A	2.5 mm²	20 m	0.10 %	PASS (<= 4%)

IEC 60364-5-52 / BS 7671 limits: Max 3% for lighting circuits, Max 5% for general power & motor loads.Total Circuits Checked: 103

E2E MultiUser Test Project 1787123629160

SHORT-CIRCUIT FAULT ANALYSIS SCHEDULE

Prepared in accordance with IEC 60364 & BS 7671 Electrical Regulations

MCGI GLASS

Ref: PRJ-921988
Date: 2026-08-19

Short-Circuit Fault Analysis Schedule

Standard: IEC 60909 / IEC 60076
3Φ Symmetrical Isc26.24 kA
2Φ Phase-to-Phase Isc22.73 kA
Peak Dynamic Stress (Ip)52.49 kA
Secondary Fault Level18.18 MVA

#	Feeder / Panel Bus	Building	Floor	Type	Cable	3Φ Isc (kA)	2Φ Isc (kA)	Breaker Icu	Protection Margin
1	⚡tower A – Main Incomer (MDB Bus)	tower A	MDB	INCOMER	Busbar	20.99 kA	18.18 kA	65 kA	🛡SAFE(Icu ≥ Isc)
2	⚡F3 – test	tower A	F3	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
3	⚡F3 – lux1	tower A	F3	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
4	⚡F3 – test	tower A	F3	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
5	⚡F3 – test	tower A	F3	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
6	⚡F3 – test	tower A	F3	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
7	⚡F3 – lux1	tower A	F3	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
8	⚡F3 – test	tower A	F3	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
9	⚡F3 – test	tower A	F3	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
10	⚡F4 – test	tower A	F4	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)

#	Feeder / Panel Bus	Building	Floor	Type	Cable	3Φ Isc (kA)	2Φ Isc (kA)	Breaker Icu	Protection Margin
11	⚡F4 – lux1	tower A	F4	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
12	⚡F4 – test	tower A	F4	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
13	⚡F4 – test	tower A	F4	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
14	⚡F4 – test	tower A	F4	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
15	⚡F4 – lux1	tower A	F4	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
16	⚡F4 – test	tower A	F4	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
17	⚡F4 – test	tower A	F4	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
18	⚡F5 – test	tower A	F5	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
19	⚡F5 – lux1	tower A	F5	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
20	⚡F5 – test	tower A	F5	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
21	⚡F5 – test	tower A	F5	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
22	⚡F5 – test	tower A	F5	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
23	⚡F5 – lux1	tower A	F5	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
24	⚡F5 – test	tower A	F5	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
25	⚡F5 – test	tower A	F5	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
26	⚡F6 – test	tower A	F6	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
27	⚡F6 – lux1	tower A	F6	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
28	⚡F6 – test	tower A	F6	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
29	⚡F6 – test	tower A	F6	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
30	⚡F6 – test	tower A	F6	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
31	⚡F6 – test	tower A	F6	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
32	⚡F6 – test	tower A	F6	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
33	⚡F6 – lux1	tower A	F6	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
34	⚡F9 – test	tower A	F9	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
35	⚡F9 – lux1	tower A	F9	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
36	⚡F9 – test	tower A	F9	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
37	⚡F9 – test	tower A	F9	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
38	⚡F9 – test	tower A	F9	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
39	⚡F9 – lux1	tower A	F9	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
40	⚡F9 – test	tower A	F9	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)

#	Feeder / Panel Bus	Building	Floor	Type	Cable	3Φ Isc (kA)	2Φ Isc (kA)	Breaker Icu	Protection Margin
41	⚡F9 – test	tower A	F9	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
42	⚡F10 – SMDB	tower A	F10	SMDB	50 mm²	7.81 kA	6.76 kA	36 kA	🛡SAFE(Icu ≥ Isc)
43	⚡F2 – SMDB	tower A	F2	SMDB	50 mm²	13.04 kA	11.29 kA	36 kA	🛡SAFE(Icu ≥ Isc)
44	⚡F7 – SMDB	tower A	F7	SMDB	50 mm²	11.57 kA	10.02 kA	36 kA	🛡SAFE(Icu ≥ Isc)
45	⚡F8 – SMDB	tower A	F8	SMDB	50 mm²	19.41 kA	16.81 kA	36 kA	🛡SAFE(Icu ≥ Isc)
46	⚡F1 – test	tower A	F1	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
47	⚡F1 – lux1	tower A	F1	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
48	⚡F1 – test	tower A	F1	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
49	⚡F1 – test	tower A	F1	APARTMENT	2.5 mm²	0.64 kA	0.55 kA	10 kA	🛡SAFE(Icu ≥ Isc)
50	⚡F10 – test	tower A	F10	APARTMENT	1.5 mm²	0.71 kA	0.61 kA	10 kA	🛡SAFE(Icu ≥ Isc)
51	⚡F10 – lux1	tower A	F10	APARTMENT	1.5 mm²	0.92 kA	0.80 kA	10 kA	🛡SAFE(Icu ≥ Isc)
52	⚡F10 – test	tower A	F10	APARTMENT	2.5 mm²	0.49 kA	0.42 kA	10 kA	🛡SAFE(Icu ≥ Isc)
53	⚡F10 – test	tower A	F10	APARTMENT	2.5 mm²	0.47 kA	0.41 kA	10 kA	🛡SAFE(Icu ≥ Isc)
54	⚡F10 – lux1	tower A	F10	APARTMENT	1.5 mm²	0.28 kA	0.24 kA	10 kA	🛡SAFE(Icu ≥ Isc)
55	⚡F10 – test	tower A	F10	APARTMENT	1.5 mm²	0.28 kA	0.24 kA	10 kA	🛡SAFE(Icu ≥ Isc)
56	⚡F10 – test	tower A	F10	APARTMENT	4 mm²	0.36 kA	0.31 kA	10 kA	🛡SAFE(Icu ≥ Isc)
57	⚡F10 – test	tower A	F10	APARTMENT	4 mm²	0.36 kA	0.31 kA	10 kA	🛡SAFE(Icu ≥ Isc)
58	⚡F2 – test	tower A	F2	APARTMENT	1.5 mm²	0.74 kA	0.64 kA	10 kA	🛡SAFE(Icu ≥ Isc)
59	⚡F2 – lux1	tower A	F2	APARTMENT	1.5 mm²	0.97 kA	0.84 kA	10 kA	🛡SAFE(Icu ≥ Isc)
60	⚡F2 – test	tower A	F2	APARTMENT	2.5 mm²	0.50 kA	0.43 kA	10 kA	🛡SAFE(Icu ≥ Isc)
61	⚡F2 – test	tower A	F2	APARTMENT	2.5 mm²	0.49 kA	0.42 kA	10 kA	🛡SAFE(Icu ≥ Isc)
62	⚡F2 – lux1	tower A	F2	APARTMENT	1.5 mm²	0.97 kA	0.84 kA	10 kA	🛡SAFE(Icu ≥ Isc)
63	⚡F2 – test	tower A	F2	APARTMENT	2.5 mm²	0.82 kA	0.71 kA	10 kA	🛡SAFE(Icu ≥ Isc)
64	⚡F2 – test	tower A	F2	APARTMENT	1.5 mm²	0.97 kA	0.84 kA	10 kA	🛡SAFE(Icu ≥ Isc)
65	⚡F2 – test	tower A	F2	APARTMENT	2.5 mm²	0.82 kA	0.71 kA	10 kA	🛡SAFE(Icu ≥ Isc)
66	⚡F7 – lux1	tower A	F7	APARTMENT	1.5 mm²	0.38 kA	0.33 kA	10 kA	🛡SAFE(Icu ≥ Isc)
67	⚡F7 – test	tower A	F7	APARTMENT	1.5 mm²	0.74 kA	0.64 kA	10 kA	🛡SAFE(Icu ≥ Isc)
68	⚡F7 – lux1	tower A	F7	APARTMENT	1.5 mm²	0.96 kA	0.83 kA	10 kA	🛡SAFE(Icu ≥ Isc)
69	⚡F7 – test	tower A	F7	APARTMENT	2.5 mm²	0.50 kA	0.43 kA	10 kA	🛡SAFE(Icu ≥ Isc)
70	⚡F7 – test	tower A	F7	APARTMENT	2.5 mm²	0.48 kA	0.42 kA	10 kA	🛡SAFE(Icu ≥ Isc)

#	Feeder / Panel Bus	Building	Floor	Type	Cable	3Φ Isc (kA)	2Φ Isc (kA)	Breaker Icu	Protection Margin
71	⚡F7 – test	tower A	F7	APARTMENT	2.5 mm²	0.32 kA	0.28 kA	10 kA	🛡SAFE(Icu ≥ Isc)
72	⚡F7 – test	tower A	F7	APARTMENT	1.5 mm²	0.38 kA	0.33 kA	10 kA	🛡SAFE(Icu ≥ Isc)
73	⚡F7 – test	tower A	F7	APARTMENT	2.5 mm²	0.32 kA	0.28 kA	10 kA	🛡SAFE(Icu ≥ Isc)
74	⚡F8 – test	tower A	F8	APARTMENT	1.5 mm²	0.76 kA	0.66 kA	10 kA	🛡SAFE(Icu ≥ Isc)
75	⚡F8 – lux1	tower A	F8	APARTMENT	1.5 mm²	1.00 kA	0.87 kA	10 kA	🛡SAFE(Icu ≥ Isc)
76	⚡F8 – test	tower A	F8	APARTMENT	2.5 mm²	0.51 kA	0.44 kA	10 kA	🛡SAFE(Icu ≥ Isc)
77	⚡F8 – test	tower A	F8	APARTMENT	2.5 mm²	0.49 kA	0.42 kA	10 kA	🛡SAFE(Icu ≥ Isc)
78	⚡F8 – test	tower A	F8	APARTMENT	2.5 mm²	0.29 kA	0.25 kA	10 kA	🛡SAFE(Icu ≥ Isc)
79	⚡F8 – test	tower A	F8	APARTMENT	1.5 mm²	0.34 kA	0.29 kA	10 kA	🛡SAFE(Icu ≥ Isc)
80	⚡F8 – lux1	tower A	F8	APARTMENT	1.5 mm²	0.34 kA	0.29 kA	10 kA	🛡SAFE(Icu ≥ Isc)
81	⚡F8 – test	tower A	F8	APARTMENT	2.5 mm²	0.29 kA	0.25 kA	10 kA	🛡SAFE(Icu ≥ Isc)
82	⚡Tower B – Main Incomer (MDB Bus)	Tower B	MDB	INCOMER	Busbar	8.02 kA	6.95 kA	65 kA	🛡SAFE(Icu ≥ Isc)
83	⚡F6 – test	Tower B	F6	APARTMENT	2.5 mm²	0.36 kA	0.31 kA	10 kA	🛡SAFE(Icu ≥ Isc)
84	⚡F6 – test2	Tower B	F6	APARTMENT	2.5 mm²	0.36 kA	0.31 kA	10 kA	🛡SAFE(Icu ≥ Isc)
85	⚡F6 – test3	Tower B	F6	APARTMENT	2.5 mm²	0.36 kA	0.31 kA	10 kA	🛡SAFE(Icu ≥ Isc)
86	⚡F7 – test	Tower B	F7	APARTMENT	2.5 mm²	0.31 kA	0.27 kA	10 kA	🛡SAFE(Icu ≥ Isc)
87	⚡F7 – test2	Tower B	F7	APARTMENT	2.5 mm²	0.31 kA	0.27 kA	10 kA	🛡SAFE(Icu ≥ Isc)
88	⚡F7 – test3	Tower B	F7	APARTMENT	2.5 mm²	0.31 kA	0.27 kA	10 kA	🛡SAFE(Icu ≥ Isc)
89	⚡F8 – test	Tower B	F8	APARTMENT	2.5 mm²	0.28 kA	0.24 kA	10 kA	🛡SAFE(Icu ≥ Isc)
90	⚡F8 – test2	Tower B	F8	APARTMENT	2.5 mm²	0.28 kA	0.24 kA	10 kA	🛡SAFE(Icu ≥ Isc)
91	⚡F8 – test3	Tower B	F8	APARTMENT	2.5 mm²	0.28 kA	0.24 kA	10 kA	🛡SAFE(Icu ≥ Isc)
92	⚡F9 – test	Tower B	F9	APARTMENT	2.5 mm²	0.25 kA	0.22 kA	10 kA	🛡SAFE(Icu ≥ Isc)
93	⚡F9 – test2	Tower B	F9	APARTMENT	2.5 mm²	0.25 kA	0.22 kA	10 kA	🛡SAFE(Icu ≥ Isc)
94	⚡F9 – test3	Tower B	F9	APARTMENT	2.5 mm²	0.25 kA	0.22 kA	10 kA	🛡SAFE(Icu ≥ Isc)
95	⚡F4 – test	Tower B	F4	APARTMENT	2.5 mm²	0.49 kA	0.42 kA	10 kA	🛡SAFE(Icu ≥ Isc)
96	⚡F4 – test2	Tower B	F4	APARTMENT	2.5 mm²	0.49 kA	0.42 kA	10 kA	🛡SAFE(Icu ≥ Isc)
97	⚡F4 – test3	Tower B	F4	APARTMENT	2.5 mm²	0.49 kA	0.42 kA	10 kA	🛡SAFE(Icu ≥ Isc)
98	⚡F5 – test	Tower B	F5	APARTMENT	2.5 mm²	0.41 kA	0.36 kA	10 kA	🛡SAFE(Icu ≥ Isc)
99	⚡F5 – test2	Tower B	F5	APARTMENT	2.5 mm²	0.41 kA	0.36 kA	10 kA	🛡SAFE(Icu ≥ Isc)
100	⚡F5 – test3	Tower B	F5	APARTMENT	2.5 mm²	0.41 kA	0.36 kA	10 kA	🛡SAFE(Icu ≥ Isc)

#	Feeder / Panel Bus	Building	Floor	Type	Cable	3Φ Isc (kA)	2Φ Isc (kA)	Breaker Icu	Protection Margin
101	⚡F1 – test3	Tower B	F1	APARTMENT	2.5 mm²	1.13 kA	0.98 kA	10 kA	🛡SAFE(Icu ≥ Isc)
102	⚡F1 – test	Tower B	F1	APARTMENT	2.5 mm²	1.13 kA	0.98 kA	10 kA	🛡SAFE(Icu ≥ Isc)
103	⚡F1 – test2	Tower B	F1	APARTMENT	2.5 mm²	1.13 kA	0.98 kA	10 kA	🛡SAFE(Icu ≥ Isc)
104	⚡F2 – test	Tower B	F2	APARTMENT	2.5 mm²	0.79 kA	0.68 kA	10 kA	🛡SAFE(Icu ≥ Isc)
105	⚡F2 – test2	Tower B	F2	APARTMENT	2.5 mm²	0.79 kA	0.68 kA	10 kA	🛡SAFE(Icu ≥ Isc)
106	⚡F2 – test3	Tower B	F2	APARTMENT	2.5 mm²	0.79 kA	0.68 kA	10 kA	🛡SAFE(Icu ≥ Isc)
107	⚡F3 – test	Tower B	F3	APARTMENT	2.5 mm²	0.61 kA	0.53 kA	10 kA	🛡SAFE(Icu ≥ Isc)
108	⚡F3 – test2	Tower B	F3	APARTMENT	2.5 mm²	0.61 kA	0.53 kA	10 kA	🛡SAFE(Icu ≥ Isc)
109	⚡F3 – test3	Tower B	F3	APARTMENT	2.5 mm²	0.61 kA	0.53 kA	10 kA	🛡SAFE(Icu ≥ Isc)

E2E MultiUser Test Project 1787123629160

BILL OF MATERIALS & PROCUREMENT SCHEDULE

Prepared in accordance with IEC 60364 & BS 7671 Electrical Regulations

Ref: PRJ-921988
Date: 2026-08-19

Bill of Materials (BOM) & Equipment Procurement Schedule

Preferred Brand: **MIXED**

1. Cable Drums & Total Conductor Sizing Bill of QuantitiesTotal Estimated Run: 2990 m

Cable Specification	Cores / System	Conductor Size	Connected Circuits	Total Estimated Length (m)
2C × 2.5 mm^2	2-Core (1φ)	2.5 mm^2	63	1840 m
2C × 4 mm^2	2-Core (1φ)	4 mm^2	2	110 m
4C × 1.5 mm^2	4-Core (3φ)	1.5 mm^2	38	980 m
4C × 95 mm^2	4-Core (3φ)	95 mm^2	1	20 m
4C × 240 mm^2	4-Core (3φ)	240 mm^2	2	40 m
Total Aggregated Cables			103 circuits	2990 m

2. Protective Switchgear & Circuit Breakers BOQ

Total Units: **109**

Rating (In)	Category	Poles	Model & Manufacturer	Sourcing Status	Quantity
16A	MCB	3P	Schneider Acti9 iC60 iC60N 3P 16A C	Catalog Match	38
25A	MCB	1P	Schneider Acti9 iC60 iC60N 1P 25A C	Catalog Match	65
160A	MCCB	3P	Schneider ComPacT NSX160 NSX160N 125A MicroLogic 2.2	Catalog Match	4
250A	MCCB	3P	Schneider ComPacT NSX250 NSX250F MicroLogic 2.2	Catalog Match	1
1000A	ACB	3P	Schneider Masterpact MTZ1 MTZ1 H1 1000	Catalog Match	1

3. Procurement Annex & Engineering Specifications for Purchase Orders

Minimum breaking capacity (Icu), trip unit specifications, and IEC standard compliance required when sourcing:

16A MCB (3P) Qty: 38
Min Breaking Cap: **10 kA**
Standard: **IEC 60898-1 / IEC 60947-2**
Trip Unit: Thermal-Magnetic (IEC 60947-2 TMD)
25A MCB (1P) Qty: 65
Min Breaking Cap: **10 kA**
Standard: **IEC 60898-1 / IEC 60947-2**
Trip Unit: Thermal-Magnetic (IEC 60947-2 TMD)
160A MCCB (3P) Qty: 4
Min Breaking Cap: **36 kA**
Standard: **IEC 60947-2**
Trip Unit: Electronic LSI (Adjustable Ir, Isd, tsd)
250A MCCB (3P) Qty: 1
Min Breaking Cap: **36 kA**
Standard: **IEC 60947-2**
Trip Unit: Electronic LSI (Adjustable Ir, Isd, tsd)
1000A ACB (3P) Qty: 1
Min Breaking Cap: **65 kA**
Standard: **IEC 60947-2**
Trip Unit: Electronic LSI / LSIG (Adjustable Ir, Isd, tsd, Ii)